

Information Security

Spring Semester 2022 ([252-0211-00L](#))

Overview

Lecturers:

Part 1: [Prof. Dr. Dennis Hofheinz](#) and [Prof. Dr. Kenneth Paterson](#)

Part 2: [Dr. Srđan Krstić](#) and [Dr. Jorge Luis Toro Pozo](#)

Assistants:

Part 1: [Varun Maram](#) (Head TA), [Julia Kastner](#), [Roman Langrehr](#), and [Dr. Fernando Virdia](#)

Part 2: [Mihael Liskij](#) (Head TA), [Sofia Giampietro](#), and [Xenia Hofmeier](#)

Course material:

Published on [Moodle](#) which also includes a discussion forum

Lectures:

Thursday 14-16 in [HG E 7](#)

Friday 14-16 in [HG E 7](#)

Links to the live stream and recordings are available on Moodle.

Exercises:

Wednesday 16-19 in [HG D 7.2](#)

Thursday 16-19 in [CAB G 61](#)

Links to the live stream and recordings are available on Moodle.

Credits: 8 ECTS (4V + 3U)

Requirements: None

Language: English

Description

This course provides an introduction to Information Security. The focus is on fundamental concepts and models, basic cryptography, protocols and system security, and privacy and data protection. While the emphasis is on foundations, case studies will be given that examine different realizations of these ideas in practice.

Exercise info

Part I

- The exercise questions will be published approximately one week before each exercise session.
- Please also note that a particular week's exercises are related to the content covered by the **preceding** week's lectures.
- Students are encouraged to work on the exercises and **attempt to solve the questions first on their own**.
- Master solutions will published a few days later.
- Students are encouraged to **study the master solutions before the exercise session**.
- In the exercise session, the TAs will explain the exercises, discuss alternative solutions, and answer the questions of the students.
- In the exercise session, it will be assumed that the students are already familiar with the questions and the master solutions.

- The exercise sessions are the **primary way of asking questions**. If a student cannot attend the exercise session, or something is not clear after the exercise session, questions can also be asked in the Moodle discussion forum.
- In the discussion forum, **students are encouraged to answer the questions of other students**. TAs will confirm correct answers and reply to any unanswered questions.
- Exercises are not graded, but **working on them** and **attending the exercise sessions** is **highly recommended**.
- The exam questions will be similar to the exercise questions. Exam questions from previous years will be part of exercises.

Part II

New assignments will be published every week (except the last one) **on Wednesday**, the contents of which will be covered in lectures that week, while the solutions will be presented in the following week.

You can hand in your solutions to the assignments via Moodle, which the teaching assistants will examine ahead of the exercise session. The feedback will then be incorporated into the exercise session without revealing your identity. (You may still choose to do so yourself, if you want.)

In order for the feedback to be incorporated during the exercise session in a timely manner, solutions should be submitted by **23:59 on Tuesday**, the week after the assignment was published.

During the exercise session, the teaching assistants will explain the correct solutions, point out common mistakes in the received submissions, and answer any questions regarding the assignments. Students who did not submit, or submitted their solutions late are still welcome to attend the session and ask questions. Master solutions will be published after the exercise session on Wednesday.

Resources

Literature

- Alfred J. Menezes, Paul C. van Oorschot, and Scott A. Vanstone: *Handbook of Applied Cryptography*, CRC Press, 1996 (available online).
- Dieter Gollmann: *Computer Security*, Wiley, 2000.
- Matt Bishop: *Computer Security: Art and Science*, Addison-Wesley, 2002 (available online for ETH members).
- Jonathan Katz and Yehuda Lindell, *Introduction to Modern Cryptography*, Chapman & Hall, 2008
- Charlie Kaufman, Rhadia Perlman, and Mike Speciner, *Network Security: Private Communication in a Public World*, 2nd Edition, 2002.
- William Stallings: *Cryptography and Network Security: Principles and Practice*, 3rd Edition, Prentice Hall, 2002.
- William Stallings: *Network Security Essentials: Applications and Standards*, 2nd Edition, Pearson, 2003.
- Ken Thompson: *Reflections on trusting trust* (available online).
- Wenbo Mao: *Modern Cryptography: Theory & Practice*, Prentice Hall, 2004.